PROJECT

Create a network with 2 routers connected by a serial link. Configure OSPF routing between them. Add 2 switches with VLANs (VLAN 10 and VLAN 20) to each router. Setup a DHCP server on Router 1 to assign IP addresses to both VLANs. Use ping and tracert commands to test connectivity between all devices. help me with this project stepwise

Objectives:

- To configure OSPF (Open Shortest Path First) dynamic routing between two routers connected via a serial link.
- To implement VLAN segmentation (VLAN 10 and VLAN 20) on switches connected to each router.
- To set up a centralized DHCP server on Router 1 to automatically assign IP addresses to all devices across both routers.
- To verify full network connectivity using ping and tracert commands between all devices.

Software Required:

- Cisco Packet Tracer

Theory:

OSPF (Open Shortest Path First)

OSPF is a link-state dynamic routing protocol used within a single Autonomous System (AS). It operates at the Network Layer (Layer 3) of the OSI model and uses IP protocol number 89. OSPF routers exchange Link State Advertisements (LSAs) to build a complete map of the network topology and use Dijkstra's algorithm to compute the shortest path to each destination.

- Works at Network Layer (Layer 3)
- Uses router-id to uniquely identify each router
- Routers form adjacency and reach FULL state before exchanging routes
- Passive interfaces prevent OSPF hello packets on LAN-facing ports
- Uses wildcard masks to define networks (inverse of subnet mask)

VLAN (Virtual Local Area Network)

A VLAN is a logical grouping of network devices that separates broadcast domains on a single physical switch. VLANs improve network security, performance, and management by isolating traffic between different groups even when they share the same physical infrastructure.

- Operates at Data Link Layer (Layer 2)
- Traffic between VLANs requires a Layer 3 device (router or Layer 3 switch)
- Trunk ports carry multiple VLANs using 802.1Q tagging
- Access ports carry traffic for a single VLAN to end devices

Router-on-a-Stick

Router-on-a-Stick is a technique where a single physical router interface is divided into multiple logical subinterfaces, each associated with a different VLAN. This allows the router to route traffic between VLANs without requiring multiple physical interfaces.

- One physical interface handles multiple VLANs
- Each subinterface is assigned a unique VLAN ID and IP address
- Encapsulation dot1Q tags frames with the correct VLAN ID

DHCP (Dynamic Host Configuration Protocol)

DHCP is a network protocol that automatically assigns IP addresses and other network configuration parameters to devices on a network. A DHCP server maintains a pool of available addresses and leases them to clients upon request.

- Eliminates manual IP configuration on end devices
- Excluded addresses reserve specific IPs for routers and servers
- DHCP relay (ip helper-address) forwards DHCP requests across different subnets
- Each VLAN requires a separate DHCP pool with matching network and gateway

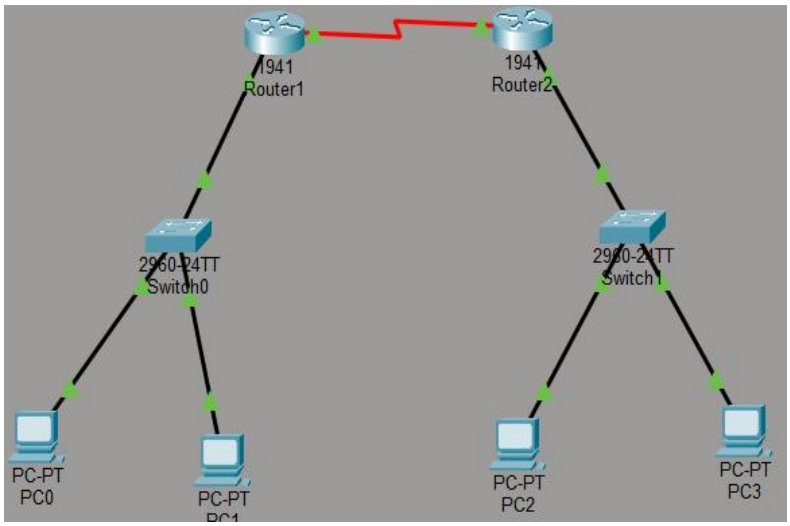
Serial WAN Link

A serial link is a point-to-point WAN connection between two routers. One end is the DCE (Data Communications Equipment) which provides the clock signal, and the other is the DTE (Data Terminal Equipment). A /30 subnet is used because only two usable IP addresses are needed.

- DCE side sets clock rate for synchronization
- /30 subnet (255.255.255.252) provides exactly 2 usable IPs
- Serial interfaces require the HWIC-2T module on Cisco 1941 routers

<u>Network Topology:</u>

The network consists of two Cisco 1941 routers connected via a serial link. Router 1 (R1) is connected to Switch0 which serves PC0 (VLAN 10) and PC1 (VLAN 20). Router 2 (R2) is connected to Switch1 which serves PC2 (VLAN 10) and PC3 (VLAN 20). R1 acts as the central DHCP server for all four subnets while R2 uses DHCP relay to forward requests from its clients to R1.



Process:

A. Build the Topology

i. Open Cisco Packet Tracer and create a new project
ii. Add 2x Router 1941, 2x Switch 2960-24TT, and 4x PC-PT
iii. Install HWIC-2T serial module on both routers (Physical tab, Power OFF, drag module, Power ON)
iv. Connect devices using Copper Straight-Through cables for PC-Switch and Switch-Router links
v. Connect routers using Serial DCE cable on Serial0/1/0 ports

B. Configure Switch0 and Switch1

vi. Create VLAN 10 and VLAN 20 in the VLAN database
vii. Set Fa0/3 as trunk port allowing VLANs 1, 10, and 20
viii. Set Fa0/1 as access port for VLAN 10
ix. Set Fa0/2 as access port for VLAN 20

C. Configure Router 1 (R1)

x. Assign IP 172.16.0.1/30 to Serial0/1/0 with clock rate 64000
xi. Create subinterfaces Gi0/1.10 and Gi0/1.20 with dot1Q encapsulation
xii. Configure DHCP pools for all four subnets
xiii. Enable OSPF process 1 with router-id 1.1.1.1 in Area 0

D. Configure Router 2 (R2)

xiv. Assign IP 172.16.0.2/30 to Serial0/1/0 (no clock rate)
xv. Create subinterfaces Gi0/1.10 and Gi0/1.20 with dot1Q encapsulation
xvi. Add ip helper-address 172.16.0.1 on both subinterfaces for DHCP relay
xvii. Enable OSPF process 1 with router-id 2.2.2.2 in Area 0

E. Set PCs to DHCP and Test

xviii. Set each PC to DHCP via Desktop > IP Configuration > DHCP
xix. Verify IP assignment using ipconfig on each PC
xx. Test connectivity using ping and tracert commands

Configuration Table:

Router Interface Configuration

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router1 (R1)	Serial0/1/0	172.16.0.1	255.255.255.252	N/A

Router1 (R1)	GigabitEthernet0/1.10	192.168.10.1	255.255.255.0	N/A
Router1 (R1)	GigabitEthernet0/1.20	192.168.20.1	255.255.255.0	N/A
Router2 (R2)	Serial0/1/0	172.16.0.2	255.255.255.252	N/A
Router2 (R2)	GigabitEthernet0/1.10	10.10.10.1	255.255.255.0	N/A
Router2 (R2)	GigabitEthernet0/1.20	10.10.20.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.10.11 (DHCP)	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.20.11 (DHCP)	255.255.255.0	192.168.20.1
PC2	FastEthernet0	10.10.10.11 (DHCP)	255.255.255.0	10.10.10.1
PC3	FastEthernet0	10.10.20.11 (DHCP)	255.255.255.0	10.10.20.1

Switch Port Configuration

Device	Interface	Mode	VLAN
Switch0 (SW0)	FastEthernet0/1	Access	VLAN 10
Switch0 (SW0)	FastEthernet0/2	Access	VLAN 20
Switch0 (SW0)	FastEthernet0/3	Trunk	1, 10, 20
Switch1 (SW1)	FastEthernet0/1	Access	VLAN 10
Switch1 (SW1)	FastEthernet0/2	Access	VLAN 20
Switch1 (SW1)	FastEthernet0/3	Trunk	1, 10, 20

DHCP Pool Configuration (Router 1)

Pool Name	Network	Default Gateway	DNS Server
VLAN10_R1	192.168.10.0/24	192.168.10.1	8.8.8.8
VLAN20_R1	192.168.20.0/24	192.168.20.1	8.8.8.8
VLAN10_R2	10.10.10.0/24	10.10.10.1	8.8.8.8
VLAN20_R2	10.10.20.0/24	10.10.20.1	8.8.8.8

OSPF Configuration

Device	Router-ID	Networks Advertised	Area
Router1 (R1)	1.1.1.1	172.16.0.0/30, 192.168.10.0/24, 192.168.20.0/24	Area 0
Router2 (R2)	2.2.2.2	172.16.0.0/30, 10.10.10.0/24, 10.10.20.0/24	Area 0

Configuration Commands:

Switch0 and Switch1 Configuration:

```
enable
configure terminal
hostname SW0
vlan 10
 name VLAN10
vlan 20
 name VLAN20
exit
interface FastEthernet0/3
 switchport mode trunk
 switchport trunk allowed vlan 1,10,20
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/2
 switchport mode access
 switchport access vlan 20
end
write memory
```

Router 1 Configuration:

```
enable
configure terminal
hostname R1
interface Serial0/1/0
 ip address 172.16.0.1 255.255.255.252
 clock rate 64000
 no shutdown
interface GigabitEthernet0/1
 no ip address
 no shutdown
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 10.10.10.1 10.10.10.10
ip dhcp excluded-address 10.10.20.1 10.10.20.10
ip dhcp pool VLAN10_R1
```

```
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20_R1
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool VLAN10_R2
 network 10.10.10.0 255.255.255.0
 default-router 10.10.10.1
 dns-server 8.8.8.8
ip dhcp pool VLAN20_R2
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.1
 dns-server 8.8.8.8
router ospf 1
 router-id 1.1.1.1
 network 172.16.0.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 passive-interface GigabitEthernet0/1.10
 passive-interface GigabitEthernet0/1.20
end
write memory
```

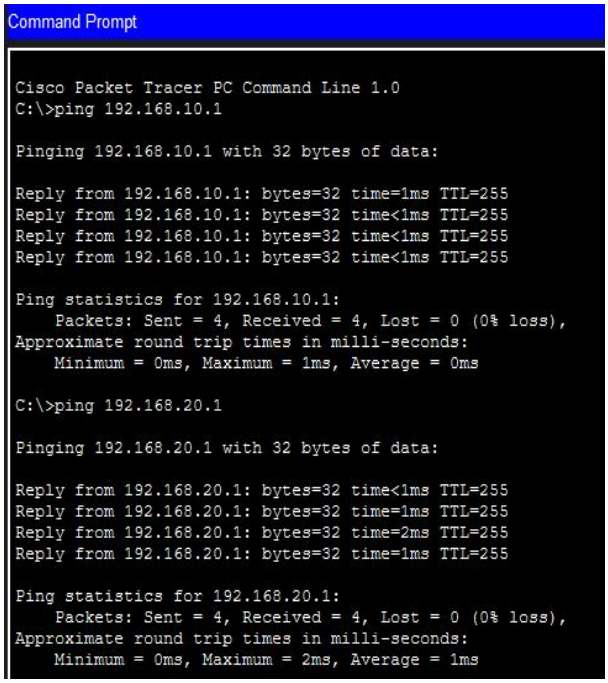
Router 2 Configuration:

```
enable
configure terminal
hostname R2
interface Serial0/1/0
 ip address 172.16.0.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 no ip address
 no shutdown
interface GigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 10.10.10.1 255.255.255.0
 ip helper-address 172.16.0.1
 no shutdown
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 10.10.20.1 255.255.255.0
 ip helper-address 172.16.0.1
 no shutdown
router ospf 1
 router-id 2.2.2.2
 network 172.16.0.0 0.0.0.3 area 0
```

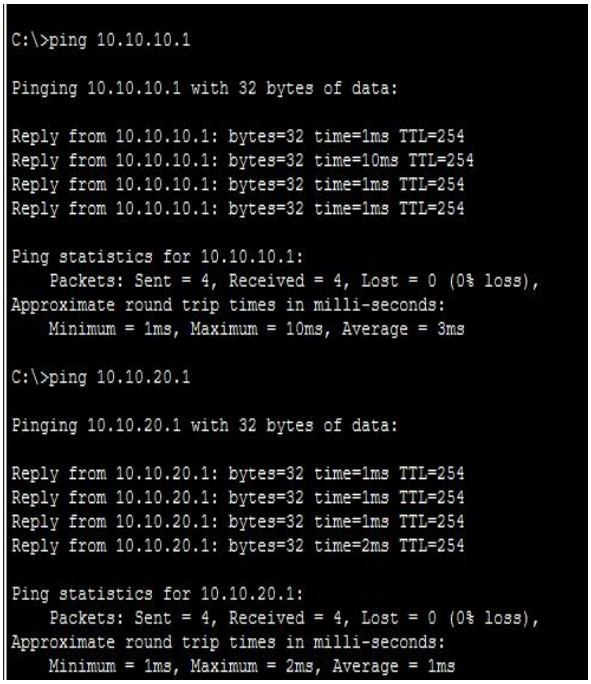
```
 network 10.10.10.0 0.0.0.255 area 0
 network 10.10.20.0 0.0.0.255 area 0
 passive-interface GigabitEthernet0/1.10
 passive-interface GigabitEthernet0/1.20
end
write memory
```

Result:

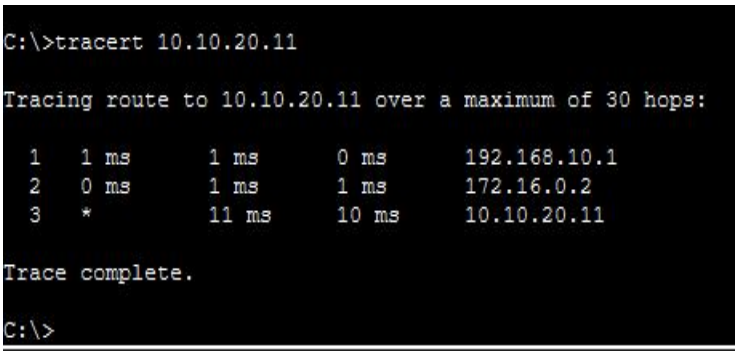
The network configuration was completed successfully. OSPF adjacency between Router 1 and Router 2 reached FULL state. All four PCs received correct IP addresses automatically from the DHCP server on Router 1. Ping tests showed 0% packet loss and tracert confirmed correct 2-hop routing across the serial link.



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=2ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```



```
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time=1ms TTL=254
Reply from 10.10.10.1: bytes=32 time=10ms TTL=254
Reply from 10.10.10.1: bytes=32 time=1ms TTL=254
Reply from 10.10.10.1: bytes=32 time=1ms TTL=254

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>ping 10.10.20.1

Pinging 10.10.20.1 with 32 bytes of data:

Reply from 10.10.20.1: bytes=32 time=1ms TTL=254
Reply from 10.10.20.1: bytes=32 time=1ms TTL=254
Reply from 10.10.20.1: bytes=32 time=1ms TTL=254
Reply from 10.10.20.1: bytes=32 time=2ms TTL=254

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```



```
C:\>tracert 10.10.20.11

Tracing route to 10.10.20.11 over a maximum of 30 hops:

  1   1 ms      1 ms      0 ms      192.168.10.1
  2   0 ms      1 ms      1 ms      172.16.0.2
  3   *         11 ms     10 ms     10.10.20.11

Trace complete.

C:\>
```

Fig: ping and tracert from pc0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.20.1

Pinging 10.10.20.1 with 32 bytes of data:

Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time=1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=2ms TTL=254
Reply from 192.168.10.1: bytes=32 time=1ms TTL=254
Reply from 192.168.10.1: bytes=32 time=2ms TTL=254
Reply from 192.168.10.1: bytes=32 time=29ms TTL=254

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 29ms, Average = 8ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=2ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\>tracert 192.168.10.1

Tracing route to 192.168.10.1 over a maximum of 30 hops:

  1   0 ms      0 ms      1 ms      10.10.10.1
  2   1 ms      0 ms      1 ms      192.168.10.1

Trace complete.

C:\>
```

Fig: ping and tracert from pc2

Discussion and Conclusion:

In this project, a network was built using two routers, two switches, and four PCs in Cisco Packet Tracer. VLAN 10 and VLAN 20 were created on both switches using the Router-on-a-Stick technique, where subinterfaces on each router handled inter-VLAN routing. OSPF was configured on both routers in Area 0, and the adjacency successfully reached FULL state, allowing routes to be exchanged automatically between all subnets.

The DHCP server on Router 1 assigned IP addresses to all four PCs across both routers. The ip helper-address command on Router 2 forwarded DHCP requests from PC2 and PC3 to Router 1, ensuring automatic IP assignment for the remote side. Connectivity was verified using ping and tracert commands, which confirmed 0% packet loss and correct 2-hop routing between the two routers.

Overall, the project was completed successfully. All objectives were achieved and the network operated with full end-to-end connectivity between all devices.